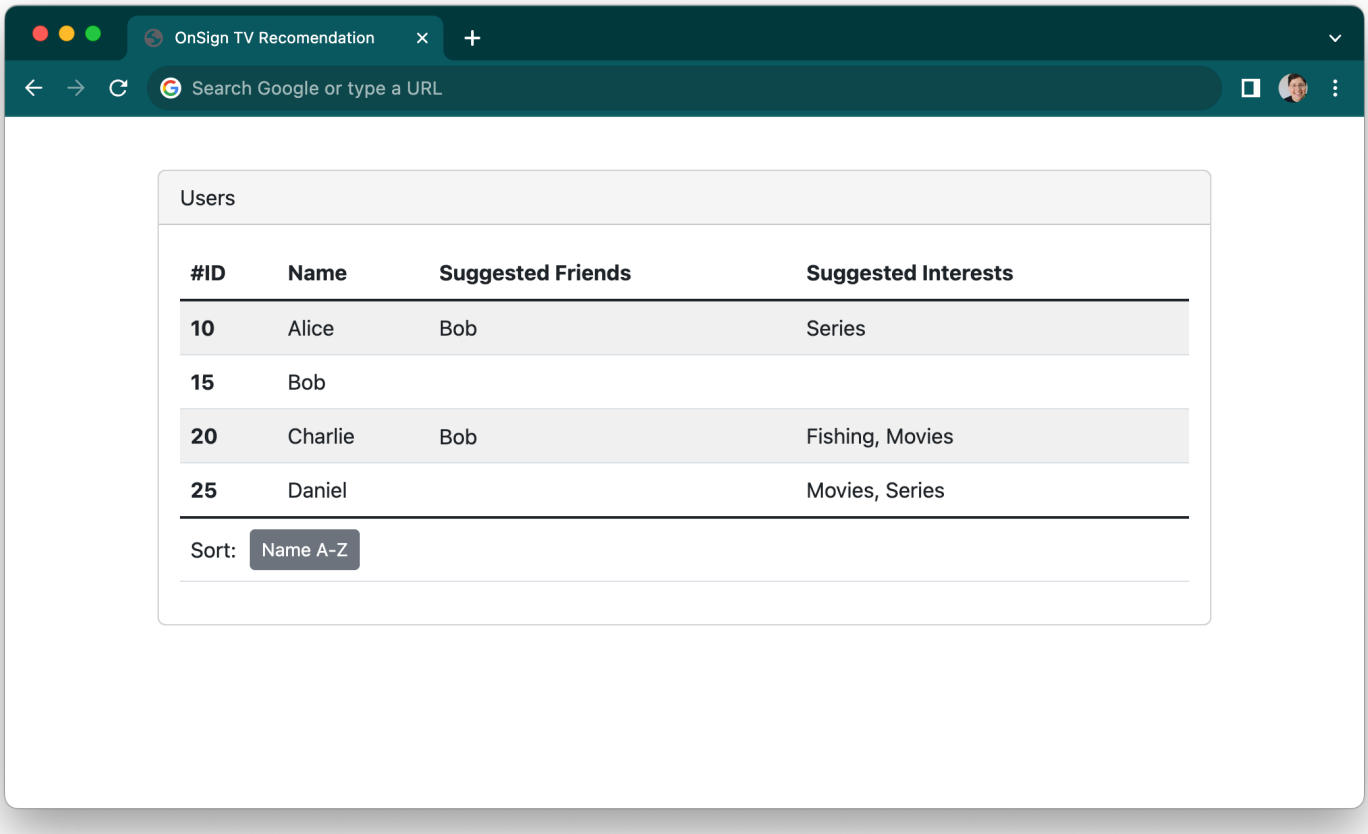
OnSign TV Dev Challenge

Hi there! Thanks for taking part in our Dev Challenge.

The goal of this challenge is to test your algorithmic ability to transform data as well as your front-end skills. This challenge shouldn't take more than a few hours to complete, depending on your experience.

We want you to create a single web page that displays a table of users, with their ids, names, some suggested friends and suggested interests, like this:



This UI is merely a suggestion. We are much more interested in how you get the data from the API and process it.

How to Get the Data

You'll get the data from this URL:

`https://api.onsign.tv/dev-challenge/?access_token=TOKEN`

Please make sure your replace the `TOKEN` parameter with the access token you received by email.

The data you'll get from that URL is in this format:

```
{
  "users": [
    { "id": 10, "name": "Alice" },
    { "id": 15, "name": "Bob" },
    { "id": 20, "name": "Charlie" },
    { "id": 25, "name": "Daniel" }
  ],
  "friends": [
    [10, 20],
    [10, 25],
    [20, 25],
    [25, 15]
  ],
  "interests": {
    "Fishing": [10, 15, 25],
    "Movies": [10, 15],
    "Series": [20, 15]
  }
}
```

You'll have an array of `"users"` that should be displayed, with their `ID`s and names.

The array of `"friends"` represents friendship connections between users. Each pair is a connection from the `"id"` in the first value to the `"id"` in the second value.

Each interest name is a key in the `"interests"` object. Each user with that interest will have their `"id"` in the corresponding array.

How to Manipulate the Data

To make things more challenging you will need to manipulate the data to extract two new pieces of information: suggested friends and suggested interests.

Suggested friends is the name of friends of friends that are not directly friends with the user.

In our example:

1. `"Alice"` is friends with `"Charlie"` and `"Daniel"`;
2. `"Charlie"` is friends with `"Daniel"`;
3. `"Daniel"` is friends with `"Bob"`;
4. The friends of friends array for `"Alice"` is `["Bob", "Daniel"]`;
5. Because `"Alice"` is already friends with `"Daniel"`, the UI only shows `"Bob"`.

Suggested interests is the list of all interests from an users friends that are not directly interests of the user.

1. In our example `"Alice"` is friends with `"Charlie"` and `"Daniel"`.
2. `"Alice"` is interested in `"Fishing"` and `"Movies"`.
3. `"Charlie"` is interested in `"Series"` and `"Daniel"` is interested in `"Fishing"`.
4. Therefore the interests of friends of Alice is `["Fishing", "Series"]`.
5. Because `"Alice"` is already interested in `"Fishing"` the UI should only show `"Series"`.

Aditional Requirements

You'll need to sort the users by name and offer a way for the user to reverse the sort order.

Please beware that the request can fail about 10% of the time. In case the status code of the request is not 200, then you'll need to wait a second and try again. You can retry up to three times before showing an error page.

Final Considerations

You can use any framework you want - or no framework at all - just make sure you choose something you are already familiar with.

We'll be evaluating your code in all senses, including names of variables and formatting, so please pay attention to those as well.

For the HTML use can use a table styled with Bootstrap: https://getbootstrap.com/docs/5.3/content/tables/

If you already know another CSS framework, use it.

Keep things simple and minimal and you'll have great success!